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Task: A Water Jug Problem

1. Problem Definition

The Water Jug Problem is a classical Artificial Intelligence problem that uses state-space search to reach a desired quantity of water. We are given two unmarked jugs: one with a capacity of 4 gallons and another with a capacity of 3 gallons. An unlimited water supply and a ground for emptying water are available.

The objective is to obtain exactly 2 gallons of water in the 4-gallon jug.

2. State Representation

The state is represented as (x, y), where:

- x = Amount of water in the 4-gallon jug.
- y = Amount of water in the 3-gallon jug.

The possible states are:

- $0 \leq x \leq 4$
- $0 \leq y \leq 3$

Initial State: (0, 0)

Goal State: (2, y), where y can be any valid quantity from 0 to 3.

3. Operators (Actions)

The following operations can be performed:

1. Fill the 4-gallon jug completely: (4, y).
2. Fill the 3-gallon jug completely: (x, 3).

3. Empty the 4-gallon jug: (0, y).
4. Empty the 3-gallon jug: (x, 0).
5. Pour water from the 4-gallon jug into the 3-gallon jug until the first jug is empty or the second jug is full.
6. Pour water from the 3-gallon jug into the 4-gallon jug until the second jug is empty or the first jug is full.

4. Step-by-Step Solution

Initially, both jugs are empty. The following sequence obtains exactly 2 gallons in the 4-gallon jug.

Step	Action Performed	4-Gallon Jug (x)	3-Gallon Jug (y)	State (x, y)
1	Initially, both jugs are empty	0	0	(0, 0)
2	Fill the 3-gallon jug	0	3	(0, 3)
3	Pour water from the 3-gallon jug into the 4-gallon jug	3	0	(3, 0)
4	Fill the 3-gallon jug again	3	3	(3, 3)
5	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full	4	2	(4, 2)
6	Empty the 4-gallon jug onto the ground	0	2	(0, 2)
7	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	2	0	(2, 0)

5. Goal State Verification

The goal is to obtain exactly 2 gallons in the 4-gallon jug.

Final State: (2, 0)

- Water in the 4-gallon jug = 2 gallons.
- Water in the 3-gallon jug = 0 gallons.

Therefore, the goal state is achieved successfully.

6. State-Space Search Representation

The Water Jug Problem can be represented as a state-space search problem in Artificial Intelligence. Each state represents the amount of water present in both jugs. The search begins from the initial state and applies valid operations until the goal state is reached.

- Initial State: (0, 0)
- Goal State: (2, 0)
- State Space: All possible valid combinations of water quantities in the two jugs.
- Operators: Fill, empty, and pour water between the jugs.
- Goal Test: Check whether the 4-gallon jug contains exactly 2 gallons.
- Path Cost: Each operation can be assigned a cost of 1. The total path cost is the number of operations performed.

7.State Transition Table

A state transition occurs when an action changes the quantity of water in one or both jugs.

Current State	Action	Next State
(0, 0)	Fill the 3-gallon jug	(0, 3)
(0, 3)	Pour into the 4-gallon jug	(3, 0)
(3, 0)	Fill the 3-gallon jug	(3, 3)
(3, 3)	Pour into the 4-gallon jug until it is full	(4, 2)
(4, 2)	Empty the 4-gallon jug	(0, 2)
(0, 2)	Pour the remaining water into the 4gallon jug	(2, 0)

The final state (2, 0) satisfies the goal condition.

8.Conclusion

The Water Jug Problem demonstrates how state-space search can be used to solve problems in Artificial Intelligence. By applying valid operations such as filling, emptying, and pouring water between jugs, we move from the initial state (0, 0) to the goal state (2, 0). Thus, exactly 2 gallons of water are obtained in the 4-gallon jug without using any measuring markings.